

Nash equilibria of games with generalized complementarities

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Abstract

A fundamental result on Nash equilibria of games with strategic complementarities is that for a quasisupermodular game, all Nash equilibria form a nonempty complete lattice. In particular, there exist largest and least Nash equilibria. Although not a quasisupermodular game, a pure Bertrand game also exhibits strategic complementarities. We show that a generalized Bertrand game has a least Nash equilibrium. We also consider an extension of quasisupermodular games and prove that the set of Nash equilibria is a nonempty complete lattice. Loosely, our observation is that a half of the single crossing property suffices for such structure results on Nash equilibria.

Keywords— Complementarities, Quasisupermodularity, Nash equilibrium, Complete lattice, Bertrand games

1 Introduction

Strategic complementarities are ubiquitous in various domains such as the oligopoly pricing and coordination games. For instance, as Samuelson [Sam74, p.1255] puts it, “tea and lemon are complements, because tea with lemon makes up our desired brew.” A joint increase of tea and lemon gives the consumer a benefit exceeding the sum of benefits gained by increasing them separately. Strategic complementarities intricately shape decision-making processes in competitive environments.

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Supermodular games ([Top98, p.179]) form a frequently used class of games with strategic complementarities. Zhou [Zho94, Thm. 2] proves that for a supermodular game where every strategy set is a compact lattice, the set of Nash equilibria is a nonempty complete lattice. Zhou’s theorem enables one to find the best-case and the worst-case equilibria, and facilitates comparative statics analysis. Topkis [Top98, Thm. 4.2.1] shows the same conclusion for a supermodular game having feasible joint strategies inside Euclidean spaces. For an overview of supermodular games and their applications, we refer the reader to [Top98], [Ami05] and [Viv05].

A method to expand the application of supermodular game theory is taking monotone transformations of the payoff functions. Features of the game determined only by ordinal properties, such as the set of Nash equilibria and the set of serially undominated strategies, remain unchanged. For instance, Milgrom and Roberts [MR90] apply the bijective monotone transformation $\log : (0, +\infty) \rightarrow \mathbb{R}$ to turn a class of non-supermodular Bertrand games to supermodular games. Milgrom and Shannon give a systematic generalization of ordinally transformed supermodular games. They extend the cardinal properties in the definition of supermodular games (supermodularity and increasing differences) to ordinal properties, known as *quasisupermodularity* ([MS94, p.162]) and *single crossing condition* ([MS94, p.160]) respectively. A normal form game whose payoff functions have the two properties is called a *quasisupermodular game* in [Top98, p.179]. The class of quasisupermodular games encompasses a broader range of scenarios compared to supermodular games, since under monotone transformations, supermodular games become quasisupermodular games, but may no longer be supermodular.

Calciano [Cal10, Def. 11] introduces *g-modularity*. This notion weakens the increasing difference hypothesis, and it is also of *cardinal* nature. For a game satisfying this condition, Calciano [Cal10, Theorems 24 and 25] shows that the set of Nash equilibria is a nonempty complete lattice. Prokopovych and Yannelis introduce several *ordinal* conditions strictly weaker than the single crossing property used in [MS94]. Their seminal results [PY17, Theorems 1 and 2] study the existence of Nash equilibria for *discontinuous* games with strategic complementarities. However, both [Cal10] and [PY17] require all the strategy sets to be *chains* (such as $[0, 1]$).

We seek to find analog of their results when the strategy sets are multidimensional (such as $[0, 1]^n$) which may not be chains. We review several *ordinal* generalizations of the quasisupermodularity and the single crossing property in the literature, and use them to define classes of games with general complementarities, strictly wider than the class of quasisupermodular games. For instance, the generalized Bertrand games studied in [CKK21, Sec. 6.2] exhibit strategic complements in a weak sense, but may not be quasisupermodular games for the lack of the single crossing property. Nevertheless, we show in Lemma 6.3 that they satisfy a “half” of the single crossing condition. Our main results focus on the existence and structure

of Nash equilibria, except that in Remark 6.6 we mention briefly how our result can be combined with [CKK21] to get a monotone comparative statics result.

Organization

Section 2 is a brief summary of quasisupermodular games. In Section 3, we recall superextremal functions, a kind of extension of quasisupermodular functions introduced by LiCalzi and Veinott. In Theorem 3.2, we show the existence of Nash equilibria when the payoff functions are superextremal and satisfy a “half single crossing property”. Similarly, in Section 4 we review pseudo-supermodular functions introduced by Agliardi. Then in Theorem 4.2, we show the existence of a largest (or dually, a least) Nash equilibrium using a “half pseudo-supermodularity”. Under the full strength of the single crossing property, we show that a “half quasisupermodularity” is sufficient for the Nash equilibria to comprise a complete lattice in Theorem 5.1. Finally in Section 6, we give several applications. Especially in Section 6.3, we apply Theorem 4.2 to study Nash equilibria of generalized Bertrand games introduced by Che, Kim and Kojima. This extends their existence result for Bertrand games with finitely many players and strategies to infinite setting. We also illustrate Theorem 5.1 in the context of minimum wage standards. Technical proofs are given in Appendix A. Appendix B explains purely order theoretic versions of the main theorems, by studying the existence of maximum of superextremal functions.

2 Quasisupermodular games

We recall the definition of quasisupermodular games with emphasis on one defining condition: the single crossing property.

Recall that a *normal form game* (called a *game* for short) $(N, \{S_i\}, \{u_i\})$ is the following data:

1. a nonempty set of players N ;
2. for every $i \in N$, a nonempty set S_i of strategies of player i ; Write $S = \prod_i S_i$ for the set of joint strategies and $S_{-i} := \prod_{j \neq i} S_j$;
3. for every player $i \in N$ a payoff function $u_i : S \rightarrow \mathbb{R}$.

For a game $(N, \{S_i\}, \{u_i\})$, a joint strategy $x \in S$ is a *Nash equilibrium* if $u_i(y_i, x_{-i}) \leq u_i(x)$ for every $i \in N$ and every $y_i \in S_i$. For every player $i \in N$, the (individual) best response correspondence is

$$R_i : S \rightarrow 2^{S_i}, \quad R_i(x) = \arg \max_{y_i \in S_i} u_i(y_i, x_{-i}).$$

Since R_i factors through the natural projection $S \rightarrow S_{-i}$, it can also be written as $R_i : S_{-i} \rightarrow 2^{S_i}$. The joint best response is

$$R : S \rightarrow 2^S, \quad R(x) = \prod_{i \in N} R_i(x_{-i}).$$

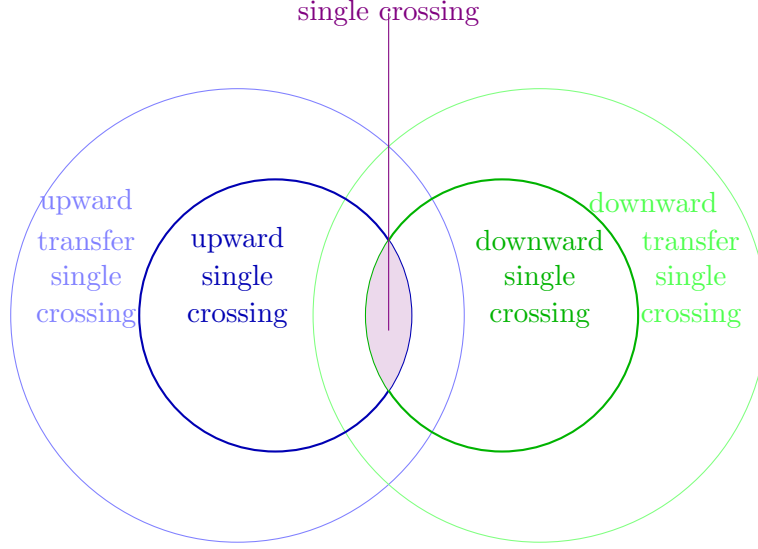
The set of Nash equilibria of the game coincides with the set $\text{Fix}(R)$ of fixed points of $R : S \rightarrow 2^S$.

A set with a partial order is called a *poset*, where a partial order on a set is an arrangement such that, for certain pairs (not all pairs) of elements, one precedes the other. Kukushkin [Kuk13b, p.1055], Prokopovych and Yannelis [PY17, p.149] introduce the following directional generalizations of the single crossing property.

Definition 2.1. Let X, T be posets. Let $f : X \times T \rightarrow \mathbb{R}$ be a function. If for any $x < x'$ in X and any $t < t'$ in T ,

1. $f(x, t) \leq f(x', t)$ implies $f(x, t') \leq f(x', t')$ (resp. $f(x, t) < f(x', t)$ implies $f(x, t') < f(x', t')$), then f is said to satisfy the *upward* (resp. *downward*) *single-crossing* property relative to (X, T) ;
2. $f(x, t) \leq f(x', t)$ (resp. $f(x, t') \geq f(x', t')$) implies the existence of $u \geq x'$ (resp. $v \leq x$) in X with $f(x, t') \leq f(u, t')$ (resp. $f(v, t) \geq f(x', t)$), then f is said to satisfy the *upward* (resp. *downward*) *transfer single-crossing* property relative to (X, T) .

In Definition 2.1, if f satisfies the upward single-crossing and the downward single-crossing properties relative to (X, T) , then Milgrom and Shannon [MS94, p.160] say that f satisfies the *single crossing property* relative to (X, T) . The single crossing property is a notion to capture the idea that “marginal returns to increasing one’s strategy rise with increases in competitors’ strategies” ([DHZ06, Footnote 6]). The upward single-crossing (resp. downward single-crossing) property implies the upward transfer single-crossing (resp. downward transfer single-crossing) property.



- Example 2.2.** 1. Let $X = T = \{0, 1\}$. Define a function $f : X \times T \rightarrow \mathbb{R}$ by $f(0, 0) = f(0, 1) = f(1, 1) = 0$ and $f(1, 0) = 1$. Then f satisfies the upward single-crossing property but not the downward single-crossing property relative to (X, T) .
2. Let $X = \{0, 1, 2\}$, $T = \{0, 1\}$. Define a function $f : X \times T \rightarrow \mathbb{R}$ by setting $f(1, 1) = -1$ and all the other values to be 0. Then f satisfies the upward transfer single-crossing property, but not the upward single-crossing property relative to (X, T) .

A poset X is called a *lattice*, if for any $x, y \in X$, their supremum $x \vee y$ and infimum $x \wedge y$ exists. A subset A of a lattice X is called a *sublattice*, if for any $x, y \in A$, one has $x \vee y, x \wedge y \in A$, i.e., A inherits a lattice structure from X . A function $f : X \rightarrow \mathbb{R}$ defined on a lattice X is called *quasisupermodular* if for all $x, y \in X$, $f(x) \geq f(x \wedge y)$ implies $f(x \vee y) \geq f(y)$, and furthermore, $f(x) > f(x \wedge y)$ implies $f(x \vee y) > f(y)$. A game $(N, \{S_i\}, \{u_i\})$ is called a *quasisupermodular game*, if for every $i \in N$, S_i is a lattice, and $u_i : S \rightarrow \mathbb{R}$ satisfies the single crossing property relative to (S_i, S_{-i}) , and moreover for every $s_{-i} \in S_{-i}$, the payoff function $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is quasisupermodular.

The *interval topology* of a poset P is the coarsest topology on P such that for every $x \in P$, $\{y \in P | y \geq x\}$ and $\{y \in P | y \leq x\}$ are closed. A lattice is called a *complete lattice*, if every nonempty subset has a supremum and an infimum. For a game $(N, \{S_i\}, \{u_i\})$, consider the following condition:

- (USC) for every $i \in N$, S_i is a lattice endowed with a compact topology

τ_i finer than the interval topology, and for every $s_{-i} \in S_{-i}$, the function $u_i(\cdot, s_{-i}) : (S_i, \tau_i) \rightarrow \mathbb{R}$ is upper semicontinuous.

In real applications, often S_i is a sublattice of a Euclidean space $\mathbb{R}^{d_i}$ and τ_i is the Euclidean topology. The reason we assume (USC) is to ensure that the best reply correspondence has *nonempty* values. By [Yu25, Corollary 6.14], quasisupermodular games satisfying (USC) retain the fundamental characteristics of supermodular games, i.e., the complete lattice structure of Nash equilibria.

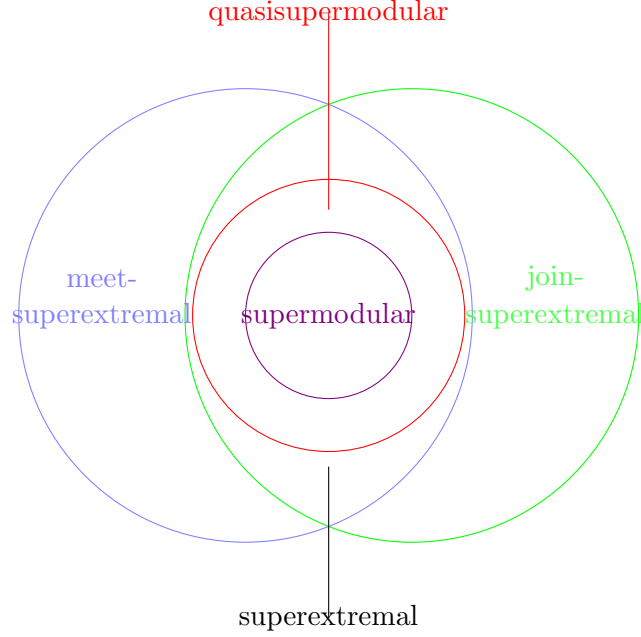
3 Existence of Nash equilibria

In Section 3, we establish the existence of Nash equilibria for a generalization of quasisupermodular games. LiCalzi and Veinott introduce a generalization of supermodular functions.

Definition 3.1 ([LV92, pp.4–5]). Let X be a lattice. A function $f : X \rightarrow \mathbb{R}$ is called

1. *meet-superextremal*, if for all $x, y \in X$, one has either $f(x \wedge y) \geq f(x)$ or $f(x \vee y) \geq f(x) \wedge f(y)$;
2. *join-superextremal*, if for all $x, y \in X$, one has either $f(x \wedge y) \geq f(x) \wedge f(y)$ or $f(x \vee y) \geq f(x)$;
3. *superextremal* if it is meet-superextremal and join-superextremal simultaneously.

Meet-superextremal roughly means that when combining two strategy profiles, at least one of the meet or join combinations improves upon an individual component—intuitively, payoffs do not dip when combining strategies. A quasisupermodular function is superextremal, but not vice versa. Shannon [Sha95] refers to a superextremal function as a *weakly quasisupermodular* function.



In Theorem 3.2, we consider the existence of Nash equilibria. The meet-superextremality assumption (resp. upward single-crossing property) is strictly weaker than the quasisupermodularity (resp. the single crossing property) used in [MS94, Thm. 12]. Be aware that there may be $i \neq i' \in N$ such that i only satisfies Condition 1 and i' only satisfies Condition 2. As a particular case, Theorem 3.2 shows the existence of Nash equilibria when every u_i is superextremal in s_i and satisfies the single crossing property in (s_i, s_{-i}) .

Theorem 3.2. *Let $(N, \{S_i\}, \{u_i\})$ be a game satisfying (USC). Assume that for every $i \in N$, either*

1. $u_i : S \rightarrow \mathbb{R}$ is upward single-crossing and for every $s_{-i} \in S_{-i}$, $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is meet-superextremal, or
2. $u_i : S \rightarrow \mathbb{R}$ is downward single-crossing and for every $s_{-i} \in S_{-i}$, $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is joint-superextremal.

Then the game admits a Nash equilibrium.

Roughly speaking, Theorem 3.2 implies that “a half of superextremality” together with “a half of single crossing property” guarantees the existence of Nash equilibria.

Remark 3.3. Consider an *aggregative game* $(N, \{S_i\}, \{u_i\})$ satisfying (USC), such that for every $i \in N$ and every $s_i \in S_i$, the function $u_i(s_i, \cdot) : S_{-i} \rightarrow \mathbb{R}$ is continuous. Acemoglu and Jensen call this game *with strategic substitutes* (instead of strategic complementarities) if u_i is supermodular in s_i and exhibits decreasing differences (instead of increasing differences) in s_i and s_{-i} . For such a game, [Jen10, Corollary 1] and [AJ13, Theorem 1] show the existence of Nash equilibria. By comparison with their result, Theorem 3.2 requires less topological continuity, and the ordinal assumptions are weaker than the supermodularity. In their proof, the aggregator plays an essential role, which may not exist neither in games considered in Theorem 3.2 nor in general supermodular games.

Remark 3.4. Kukushkin [Kuk18, Theorem 2] also proves the existence of Nash equilibria for discontinuous games with strategic complementarities. Theorem 3.2 and his result share *common assumptions* that

- every S_i is a compact lattice,
- and that every $u_i : S \rightarrow \mathbb{R}$ is downward single-crossing relative to (S_i, S_{-i}) .

The differences are as follows. He assumes further that every S_i is a *distributive* lattice, while we do not. His hypothesis [Kuk18, (5a)] is stronger than our meet-superextremality hypothesis. Kukushkin's *Q-finite deviation property* plays a role similar to our upper semicontinuity assumption, but neither implies the other.

Remark 3.5. For superextremal games satisfying certain measure theoretic hypotheses, [Bal+19, Theorem 1 and Remark 2] show the existence of the largest Nash equilibrium. For games whose payoff functions are all meet-superextremal (or all join-superextremal), Sabarwal [Sab25, Example 6] establishes monotone comparative statics.

In Theorem 3.2, the set of Nash equilibria may not have largest nor least elements, and it may not be a lattice, as a banal example shows.

Example 3.6. Let $N = \{1\}$. Let $S_1 = \{(0, 0), (1, 0), (0, 1), (1, 1), (2, 1), (1, 2), (2, 2)\}$ be a subset of $\mathbb{R}^2$. Let τ_1 be the discrete topology of S_1 . Define a function $u_1 : S_1 \rightarrow \mathbb{R}$ such that $u_1((0, 0)) = u_1((2, 2)) = 0$ and u_1 takes value 1 at other points. Then the corresponding game satisfies the condition of Theorem 3.2. The set of Nash equilibria is $S_1 \setminus \{\max S_1, \min S_1\}$, which is not a lattice. It has no largest nor least elements.

4 Largest and least Nash equilibria

A poset where any two elements are comparable is called a *chain*. For discontinuous games with strategic complementarities, a seminal result [PY17, Theorem 1] establishes

the existence of Nash equilibria provided that the strategy sets are chains. In many economics-related games, such as production games with complementary products and facility location games ([Top98, 4.4.2 and 4.4.7]), the strategy sets are multidimensional instead of chains. We generalize Prokopovych and Yannelis’s theorem by removing the chain assumption. Our result encompasses a class of games containing both quasisupermodular games and pseudo-supermodular games. We show not only the existence of a Nash equilibrium, but also that of a largest or a least one. Then we apply the result to Bertrand games in Section 6.3.

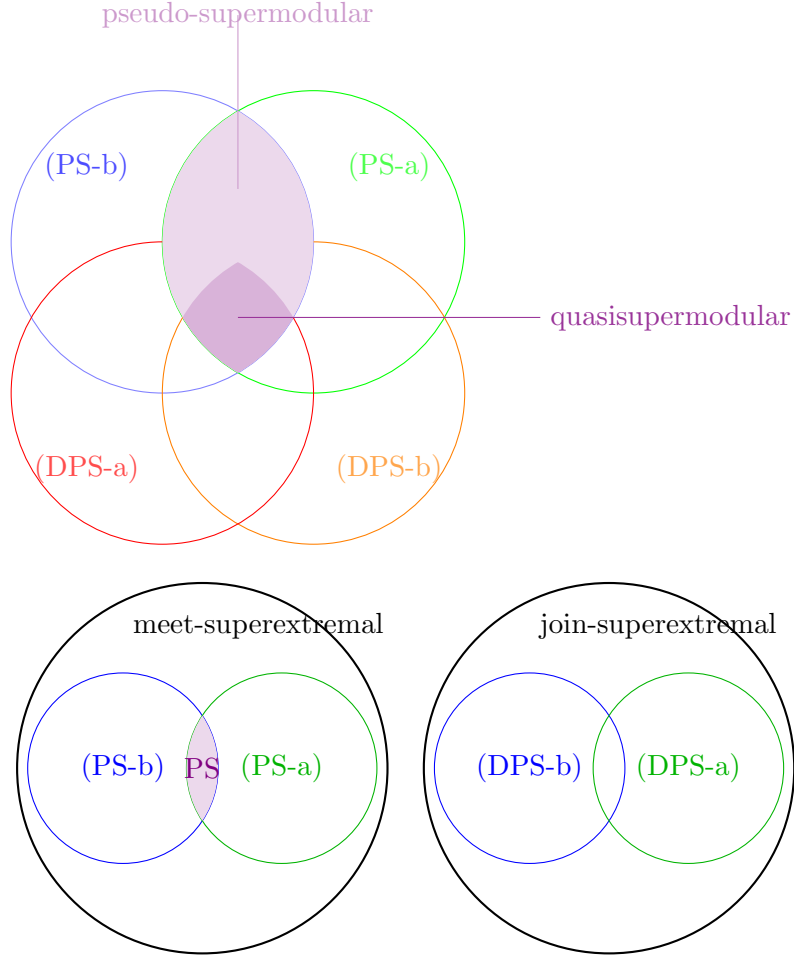
For a function $f : X \rightarrow \mathbb{R}$ on a lattice X , we review the conditions [Kuk13b, (10a), (10b), (11a), (11b)] in order.

- (PS-a) For all $x, y \in X$, $f(x \wedge y) < f(x) \vee f(y)$ implies $f(x \vee y) > f(x) \wedge f(y)$.
- (PS-b) For all $x, y \in X$, $f(x \vee y) < f(x) \wedge f(y)$ implies $f(x \wedge y) > f(x) \vee f(y)$.
- (DPS-b) For all $x, y \in X$, $f(x \wedge y) < f(x) \wedge f(y)$ implies $f(x \vee y) > f(x) \vee f(y)$.
- (DPS-a) For all $x, y \in X$, $f(x \vee y) < f(x) \vee f(y)$ implies $f(x \wedge y) > f(x) \wedge f(y)$.

Remark 4.1. Agliardi [Agl00, p.252] calls f *pseudo-supermodular* if f satisfies Conditions (PS-a) and (PS-b), so we use “PS” as short for “pseudo-supermodularity”. The four conditions are dual to each other. For example, f satisfies (DPS-b) exactly when the induced function

$$g : X^{\text{op}} \rightarrow \mathbb{R}, \quad x \mapsto f(x)$$

satisfies (PS-b). For this reason, we use labels (DPS-a) and (DPS-b), where “DPS” is short for “dual pseudo-supermodularity”. A pseudo-supermodular function may not be join-superextremal. Superextremality does not imply any of the four conditions. Moreover, $f : X \rightarrow \mathbb{R}$ satisfies the four conditions if and only if it is quasisupermodular.



Compared with Theorem 3.2, the existence of the largest Nash equilibrium in Theorem 4.2 is stronger than the mere existence of Nash equilibria. The upward transfer single-crossing condition is weaker than the upward single-crossing condition. Still, Condition (PS-b) in Theorem 4.2 is stronger than the meet-superextremality hypothesis in Theorem 3.2.

Theorem 4.2. *Let $(N, \{S_i\}, \{u_i\})$ be a game satisfying (USC). Assume that for every $i \in N$,*

1. *for every $s_{-i} \in S_{-i}$, the function $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ satisfies Condition (PS-b) (resp. Condition (DPS-b));*
2. *the payoff function $u_i : S \rightarrow \mathbb{R}$ satisfies the upward transfer single-crossing (resp. downward transfer single-crossing) property relative to (S_i, S_{-i}) .*

Then the game admits a largest (resp. least) Nash equilibrium.

Loosely speaking, Theorem 4.2 implies that “a half of pseudo-supermodularity” combined with “a half of single crossing property” ensures the existence of the largest Nash equilibrium.

Remark 4.3. Given a quasisupermodular game $(N, \{S_i\}, \{u_i\})$ satisfying (USC), Milgrom and Shannon [MS94, Theorem 12] prove that if every u_i is continuous in s_{-i} , then there exists a largest Nash equilibrium and a least Nash equilibrium, which are respectively the largest and the smallest serially undominated strategies. Intuitively, in Theorem 4.2 Assumption 1 is “a quarter of the quasisupermodularity”, and Assumption 2 is weaker than “a half of the single crossing property”. Our assumption is more general than that of Milgrom and Shannon, but our result is also weaker as we do not study the question whether the largest Nash equilibrium is the largest serially undominated strategy profile.

Remark 4.4. Agliardi calls a game $(N, \{S_i\}, \{u_i\})$ *pseudo-supermodular*, if every u_i satisfies (PS-a) and (PS-b) in s_i , and satisfies the single crossing property in (s_i, s_{-i}) as well as an extra hypothesis [Agl00, Condition (A), p.253]. She [Agl00, Proposition 4] proves that if a pseudo-supermodular game satisfies (USC) and if every u_i is continuous in s_{-i} , then there is a largest Nash equilibrium and a least Nash equilibrium which are respectively the largest and the smallest strategies which survive strict pure iterated admissibility. On the one hand, compared with her assumption, Theorem 4.2 only assumes (PS-b) and the upward transfer single-crossing property. On the other hand, her result is stronger in the following sense. We only show the existence of the largest Nash equilibrium, and we do not study the question whether it is the largest strategy profile which survives strict pure iterated admissibility.

Remark 4.5. Theorem 4.2 is a generalization of [PY17, Theorem 1] to multidimensional strategies. The authors only mention the existence of Nash equilibria, require finiteness of N and a more restrictive ordinal condition that every S_i is a *chain*. Strictly speaking, they use a weaker continuity condition. Instead of the topological upper semicontinuity of u_i in our (USC) assumption, they only assume that every u_i is transfer weakly upper continuous in s_i and upward upper semicontinuous in s_i . The existence of the largest Nash equilibrium holds even if we weaken upper semicontinuity in Theorem 4.2 to their continuity assumption. In fact, as a replacement of Lemma A.1, we can show that for every $s_{-i} \in S_{-i}$, $R_i(s_{-i})$ is nonempty and every nonempty chain inside admits an upper bound (see the proof of [Yu25, Theorem 6.11]).

5 Complete lattice structure of Nash equilibria

Vives [Viv90, Theorem 4.2 (ii)] (in a special case) and Zhou [Zho94, Theorem 2] (in the general case) prove that given a supermodular game $(N, \{S_i\}, \{u_i\})$ satisfying (USC), the Nash equilibria form a nonempty complete lattice. In particular, there exist largest and least Nash equilibria. This fundamental theorem helps to study the dynamic of Nash equilibria.

Consider a *weakly* monotone transformation which may carry different real numbers to the same value. Then the transform of a quasisupermodular game may no longer be quasisupermodular. In certain cases, such a transformation preserves only “a half of the quasisupermodularity”. We show that such a “half quasisupermodularity” of payoff functions, together with the single crossing property, is enough to guarantee that the game has a complete lattice as its set of Nash equilibria. An illustration is given in Section 6.4.

The class of games satisfying the assumptions of Theorem 5.1 is a strict superset of the union of the class of *games with strategic complementarities* in the sense of [MS94, p.175] and pseudo-supermodular games in [Agl00, Proposition 4]. Thus, Theorem 5.1 recovers partially [MS94, Theorem 12] and [Agl00, Proposition 4], namely the existence of the largest and the least Nash equilibria.

Theorem 5.1. *Let $(N, \{S_i\}, \{u_i\})$ be a game satisfying (USC). Assume that for every $i \in N$,*

1. *at least one of (PS-a) and (DPS-b) is satisfied by $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ for all $s_{-i} \in S_{-i}$;*
2. *at least one of (PS-b) and (DPS-a) is satisfied by $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ for all $s_{-i} \in S_{-i}$;*
3. *the function $u_i : S \rightarrow \mathbb{R}$ satisfies the single crossing property relative to (S_i, S_{-i}) .*

Then the set of Nash equilibria is a nonempty complete lattice.

Theorem 5.1 implies that “a half of the quasisupermodularity” and the single crossing property are sufficient to ensure that the set of equilibria forms a complete lattice.

6 Applications

We provide some applications of the results on Nash equilibria.

6.1 Multi-winner competition

Given a discontinuous game, the single crossing property may fail while the upward single-crossing property still holds. Consider a nonempty set N of competing companies seeking cooperation with another trader t . Suppose that t only trades with competitors offering the highest price. As a tie-breaking rule, we assume that if more than one competitors offer the same highest price, then each of them trades with t . For a competitor $i \in N$, let $v_i \in [0, +\infty)$ be her valuation of this cooperation. We assume that every competitor is rational, so competitor i never bids more than v_i . Her set of strategies is the compact interval $S_i := [0, v_i]$. Her payoff function is

$$u_i : S \rightarrow [0, v_i], \quad u_i(s) = \begin{cases} v_i - s_i, & s_i = \max_{j \in N} s_j, \\ 0, & s_i < \max_{j \in N} s_j. \end{cases}$$

It is upper semicontinuous in the variable s_i when $s_{-i} \in S_{-i}$ is fixed. Therefore, the game satisfies (USC) condition. Because S_i is a chain, u_i is supermodular in the variable s_i .

Take $s_i < s'_i$ in S_i and $s_{-i} < s'_{-i}$ in S_{-i} with

$$s_i < \max_{j \neq i} s_j < s'_i < \min(\max_{j \neq i} s'_j, v_i).$$

Then $u_i(s_i, s_{-i}) = 0 < v_i - s'_i = u_i(s'_i, s_{-i})$ and $u_i(s_i, s'_{-i}) = 0 = u_i(s'_i, s'_{-i})$. Thus, u_i does not satisfy the single crossing property. Nevertheless, we prove that u_i satisfies the upward single-crossing property relative to (S_i, S_{-i}) . For any $s_i < s'_i$ in S_i and $s_{-i} < s'_{-i}$ in S_{-i} with $u_i(s_i, s'_{-i}) > u_i(s'_i, s'_{-i})$, one has $s_i \geq \max_{j \neq i} s'_j$. It implies $u_i(s_i, s_{-i}) = v_i - s_i > v_i - s'_i = u_i(s'_i, s_{-i})$. Therefore, by Theorem 4.2, this game admits a largest Nash equilibrium.

6.2 Multimarket Oligopoly

We extend [PY17, Example 4] from the case of a single project to multiple projects. This model is a kind of the multimarket oligopoly studied in [Top98, 4.4.3]. Consider n partners cooperate in m projects. Each partner $i \in N = \{1, 2, \dots, n\}$ chooses the vector of effort s_i in $S_i := [0, 1]^m$, where the j -th coordinate $s_i^j \in [0, 1]$ is her effort to contribute to project j . Let $S^j := [0, 1]^n$. Given a profile $s^j := (s_1^j, \dots, s_n^j) \in S^j$ of efforts in project j , the total output of project j is $f^j(s^j) = \sum_{i \in N} s_i^j$, and partner i obtains the fraction $p_i^j(s^j)$ of the total output, where

$$p_i^j : S^j \rightarrow [0, 1], \quad s^j \mapsto \begin{cases} 1/k & s_i^j = \max_{i' \in N} s_{i'}^j \text{ and } \#\{i' \in N \mid s_{i'}^j = s_i^j\} = k \\ 0 & s_i^j < \max_{i' \in N} s_{i'}^j. \end{cases}$$

The profit is evenly shared by those exerting the highest effort. Player i 's payoff from project j is

$$u_i^j : S^j \rightarrow \mathbb{R}, \quad s^j \mapsto p_i^j(s^j)f^j(s^j) - s_i^j,$$

and her payoff from the m projects in total is

$$u_i : S \rightarrow \mathbb{R}, \quad s \mapsto \sum_{j=1}^m u_i^j(s^j).$$

This game is symmetric, i.e., exchangeable against permutations of the players.

For every $i \in N$ and every fixed $s_{-i} \in S_{-i}$, one has $u_i(s_i, s_{-i}) = \sum_{j=1}^m u_i^j(s_i^j, s_{-i}^j)$, so the function $u_i(\cdot, s_{-i}) : S_i = [0, 1]^m \rightarrow \mathbb{R}$ is separable in the sense of [Top98, p.25]. Then by [Top98, Theorem 2.6.4], this function is supermodular.

We show $u_i^j(s^j) \leq u_i^j(1, s_{-i}^j)$ for every $1 \leq j \leq m$ and every $s^j \in S^j$. As it holds when $s_i^j = 1$, one may assume $s_i^j < 1$. If $\max_{i' \neq i} s_{i'}^j < 1$, then $p_i^j(1, s_{-i}^j) = 1$ and hence

$$u_i^j(s^j) \leq f^j(s^j) - s_i^j = \sum_{i' \neq i} s_{i'}^j = f^j(1, s_{-i}^j) - 1 = u_i^j(1, s_{-i}^j).$$

If $\max_{i' \neq i} s_{i'}^j = 1$, then $p_i^j(s^j) = 0$. Let $k := \#\{i' \in N \mid s_{i'}^j = 1\}$. Then $p_i^j(1, s_{-i}^j) = 1/k$ and $\sum_{i' \in N} s_{i'}^j \geq k$. Therefore, one has

$$u_i^j(s^j) = -s_i^j \leq 0 \leq p_i^j(1, s_{-i}^j)f^j(1, s_{-i}^j) - 1 = u_i^j(1, s_{-i}^j).$$

The inequality is proved in both cases.

Let $\bar{s}_i := (1, \dots, 1) \in [0, 1]^m$ be the largest element of S_i . Thus, one has $u_i(s) \leq u_i(\bar{s}_i, s_{-i})$ for every $s \in S$, i.e., player i always has a best replay $\bar{s}_i$ whatever $s_{-i} \in S_{-i}$ is. In particular, $u_i : S \rightarrow \mathbb{R}$ satisfies the upward transfer single-crossing property relative to (S_i, S_{-i}) . It does not satisfy the upward single crossing property.

For every $i \in N$, u_i is transfer weakly upper semicontinuous and upward upper semicontinuous in s_i . Then by Remark 4.5, the game has a largest Nash equilibrium, which turns out to be $\bar{s} := (\bar{s}^j)_{i \in N}$. When $m > 1$, no S_i is a chain, in which case one cannot apply [PY17, Theorem 1].

6.3 Generalized Bertrand games

Review of Bertrand games

A Bertrand game is a pricing problem in a market with n competitors. Each competitor $i = 1, \dots, n$ produces a product still denoted by i . Assume that the

n products are substitutes for one another. They can be the same product with different labels, or different items with similar functions. Each competitor i faces the problem of determining the price $s_i \geq 0$ for product i . The demand D_i for product i depends on the price vector $s = (s_1, \dots, s_n) \in \mathbb{R}^n$. The cost in production of i is $C_i(D_i)$. Thus, the net profit for competitor i is

$$u_i(s) = s_i D_i(s) - C_i(D_i(s)).$$

Topkis [Top79, Example (c)] considers the case with linear cost, i.e., for every i , there exists a unit production cost constant c_i with $C_i(D_i) = c_i D_i$. Assuming that D_i has increasing differences in (s_i, s_j) for all $j \neq i$ [Top79, Condition (B)], Topkis verifies that such a Bertrand game is a supermodular game. Still, the only commonly studied demand functions with increasing differences [Top79, Condition (B)] are linear functions. Under the hypothesis that the C_i are convex functions, Vives [Viv90, 7.2.1] also analyzes a sufficient condition for a Bertrand game to be a supermodular game.

Milgrom and Roberts [MR90, Sec. 4 (2)] consider a Bertrand game with linear cost functions satisfying a log-increasing difference condition:

- (log-ID) the demand functions D_i are positive and continuously differentiable, and $\partial^2 \log D_i(s) / \partial s_i \partial s_j \geq 0$ for all $1 \leq j \neq i \leq n$.

They show that even if the original Bertrand game is not necessarily supermodular, the log-transformed game is supermodular. The inequality $\partial^2 \log D_i(s) / \partial s_i \partial s_j \geq 0$ in Condition (log-ID) reflects the assumption that the elasticity of demand increases with the other firms' prices. Milgrom and Shannon [MS94, p.168] extend the result of Milgrom and Roberts to the case of nonlinear cost functions. They show that under Condition (log-ID), if the cost functions are continuous, increasing and convex, then the Bertrand game is a quasisupermodular game.

Least Nash equilibria of generalized Bertrand games

In [CKK21, Sec. 6.2], the authors introduce a generalization of Bertrand games. Then in [CKK21, Corollary 4], they investigate the monotone comparative statics for this class of games. We apply Theorem 4.2 to show that a generalized Bertrand game has a least Nash equilibrium.

Definition 6.1 ([CKK21, p.29]). A game $(N, \{S_i\}, \{u_i\})$ is called a *generalized Bertrand game*, if for every $i \in N$,

- S_i is a subset of $[0, +\infty)$;
- there is a function $D_i : S \rightarrow [0, +\infty)$ and a convex function $C_i : [0, +\infty) \rightarrow [0, +\infty)$, such that $u_i(s) = s_i D_i(s) - C_i(D_i(s))$ for all $s \in S$;

- the function $D_i : S \rightarrow [0, +\infty)$ satisfies the following conditions.
 - (D1) For every $s_{-i} \in S_{-i}$, the function $D_i(\cdot, s_{-i}) : S_i \rightarrow [0, +\infty)$ is decreasing; for every $s_i \in S_i$, the function $D_i(s_i, \cdot) : S_{-i} \rightarrow [0, +\infty)$ is increasing;
 - (D2) For any $s_i < s'_i$ in S_i and $s_{-i} < s'_{-i}$ in S_{-i} with $D_i(s_i, s_{-i}) > 0$, one has $D_i(s'_i, s_{-i})/D_i(s_i, s_{-i}) \leq D_i(s'_i, s'_{-i})/D_i(s_i, s'_{-i})$.

A generalized Bertrand game represents an oligopoly game played by firms N whose products are substitutes for each other. The formula of u_i signifies that each firm $i \in N$ chooses a price $s_i \in S_i$ and sells $D_i(s)$ units of products at a cost $C_i(D_i(s))$. Assumption (D1) has a natural interpretation. When the price $s_i \in S_i$ goes up and $s_{-i} \in S_{-i}$ remains fixed, the demand $D_i(s)$ decreases. When the price $s_i \in S_i$ remains fixed and $s_{-i} \in S_{-i}$ gets higher, as the products are substitutes, the demand $D_i(s)$ increases. Assumption (D2) is equivalent to that the function $\log D_i : \{s \in S | D_i(s) > 0\} \rightarrow \mathbb{R}$ has increasing differences relative to (S_i, S_{-i}) . Thus, (D2) is weaker than Condition (log-ID). Intuitively, (D2) implies that as the price $s_{-i} \in S_{-i}$ of the other firms decreases, the effect of decreasing $s_i \in S_i$ on increasing the demand $D_i(s)$ becomes more significant.

Example 6.2. Consider a two-player game as follows. Let $N = \{1, 2\}$. Let strategy sets be $S_1 = \{1, 3\}$ and $S_2 = \{3, 4\}$. Define demand functions by

$$D_1(1, 3) = 1, D_1(1, 4) = 1, D_1(3, 3) = 1/2, D_1(3, 4) = 1, D_2 \equiv 1 - D_1.$$

Intuitively, the firm with lower price occupies the whole market demand. In the case of a tie of prices, the two firms share the demand equally. Let $c_1 = 0$, $c_2 = 2$ be unit production cost for firms 1 and 2 respectively. Then the total production cost for firm i is $C_i(D_i) = c_i D_i$. Therefore, the payoff functions are given by

$$\begin{aligned} u_1(1, 3) &= 1, u_1(1, 4) = 1, u_1(3, 3) = 1.5, u_1(3, 4) = 3, \\ u_2(1, 3) &= u_2(1, 4) = u_2(3, 4) = 0, \quad u_2(3, 3) = 0.5. \end{aligned}$$

This is a pure Bertrand game in the sense of [CKK21, Footnote 37]. By [CKK21, Lemma S3], every pure Bertrand game is a generalized Bertrand game. In this case, u_2 does not satisfy the single crossing property relative to (S_2, S_1) , since $u_2(1, 3) \leq u_2(1, 4)$ while $u_2(3, 3) > u_2(3, 4)$.

As Example 6.2 shows, a generalized Bertrand game may not be a quasisupermodular game. Loosely speaking, Lemma 6.3 proves that it satisfies a “half single crossing property”.

Lemma 6.3. *Given a generalized Bertrand game $(N, \{S_i\}, \{u_i\})$, for every $i \in N$, the payoff function $u_i : S \rightarrow \mathbb{R}$ is downward single-crossing relative to (S_i, S_{-i}) .*

Corollary 6.4. *Let $(N, \{S_i\}, \{u_i\})$ be a generalized Bertrand game. For every $i \in N$, assume that S_i is compact in $[0, +\infty)$, and $C_i : D_i(S) \rightarrow \mathbb{R}$ is continuous. Suppose that for every $s_{-i} \in S_{-i}$, the function $D_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is continuous. Then there is a least Nash equilibrium.*

Proof. For every $i \in N$, the Euclidean topology restricted to S_i is exactly the interval topology. From Lemma 6.3, $u_i : S \rightarrow \mathbb{R}$ is downward transfer single-crossing relative to (S_i, S_{-i}) . As S_i is a chain, for every $s_{-i} \in S_{-i}$, the function $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is supermodular. By assumption, it is also continuous. Thus, the game satisfies (USC). Therefore, by Theorem 4.2, this game has a least Nash equilibrium. $\square$

Remark 6.5. Every convex function is continuous on the interior of its domain (see, e.g., [FLVG01, Theorem 5.2.1]). Therefore, for every $i \in N$, the continuity hypothesis on $C_i : D_i(S) \rightarrow \mathbb{R}$ in Corollary 6.4 is automatic whenever $0 \notin D_i(S)$.

Remark 6.6. A generalized Bertrand game is *finite* if N and the S_i are finite. In this case, S and hence $D_i(S)$ are finite. Thus, $C_i : D_i(S) \rightarrow \mathbb{R}$ and $D_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ are continuous. Then by Corollary 6.4, a finite generalized Bertrand game has a least Nash equilibrium. The existence of Nash equilibria for *finite* generalized Bertrand games is covered by [CKK21, Corollary 4]. As [CKK21, p.29] remarks, finiteness is assumed only “to guarantee the existence of a Nash equilibrium in pure strategies, but otherwise plays no role.” Thereby, as Corollary 6.4 ensures the existence without the finiteness assumption, one can extend the monotone comparative statics result [CKK21, Corollary 4] to infinite generalized Bertrand games.

6.4 Guaranteed minimum income

As another application, we analyze the transform of quasisupermodular games under a particular weakly monotone transformation, namely the truncation at a constant. For a set S , a function $f : S \rightarrow \mathbb{R}$ and a constant $c \in \mathbb{R} \cup \{-\infty\}$, we call the function $\underline{f} := \max(f, c)$ the *truncation* of f at level c . For example, if S represents the strategy profiles of some competing workers, f represents the original income of a fixed worker $i_0 \in N$ and if c is a newly introduced minimum income standard, then $\underline{f}$ can be interpreted as the new income of i_0 . (Surely this is only a simplification of a real situation.) Guaranteed minimum income is a social-welfare system that no one will fall below a set income level that would allow them to live a dignified life. For the socio-economic impact of minimum income schemes, see, e.g., [FM16] and [Rai+21].

In Corollary 6.9, we show that the introduction of a minimum income standard does not change a notable feature of games with complementarities: the Nash equilibria remain a complete lattice.

Lemma 6.7. *Let $f : X \rightarrow \mathbb{R}$ be a function defined on a lattice. Let $a \in \mathbb{R}$. Let $\underline{f} : X \rightarrow \mathbb{R}$ be the truncation of f at level a . If f satisfies (PS-b) (resp. (DPS-b)), then $\underline{f}$ also satisfies (PS-b) (resp. (DPS-b)).*

Proof. We prove the case with (PS-b). The case in parentheses follows by symmetry. Let $x, y \in X$ be such that $\underline{f}(x \vee y) < \underline{f}(x) \wedge \underline{f}(y)$. Then $a < \underline{f}(x) \wedge \underline{f}(y)$, so $a < f(x) \wedge f(y)$. Thus, one has $f(x \vee y) \leq \underline{f}(x \vee y) < f(x) \wedge f(y)$. As f satisfies (PS-b), one has $f(x \wedge y) > f(x) \vee f(y) > a$. It implies $\underline{f}(x \wedge y) > \underline{f}(x) \vee \underline{f}(y)$. Therefore, $\underline{f}$ satisfies (PS-b). $\square$

Example 6.8 illustrates that the truncation of a supermodular function may not be quasisupermodular.

Example 6.8. Let $X = \{0, 1, x, y\}$, where x, y are incomparable, $x \wedge y = 0$ and $x \vee y = 1$. Then X is a lattice. The function

$$f : X \rightarrow \mathbb{R}, \quad f(0) = 1, \quad f(1) = -1, \quad f(x) = 2, \quad f(y) = -2$$

is supermodular. Its truncation at level $a = 0$ is

$$\underline{f}(0) = 1, \quad \underline{f}(1) = 0, \quad \underline{f}(x) = 2, \quad \underline{f}(y) = 0,$$

which does not satisfy (PS-a).

Corollary 6.9. *Let $(N, \{S_i\}, \{u_i\})$ be a quasisupermodular game satisfying (USC). Let $c \in \mathbb{R} \cup \{-\infty\}$. Assume that for every $i \in N$ and every $s_i \in S_i$, if there is one $s_{-i} \in S_{-i}$ with $u_i(s_i, s_{-i}) > c$, then $u_i(s_i, s'_{-i}) > c$ for all $s'_{-i} \in S_{-i}$. For each $i \in N$, let $\underline{u}_i$ be the truncation of u_i at level c . Then the set of Nash equilibria of the new game $(N, \{S_i\}, \{\underline{u}_i\})$ is a nonempty complete lattice.*

In Corollary 6.9, if c is interpreted as a minimum income standard, then the assumption on c means that whether a worker i earns more than the minimum level depends only on her own strategy s_i but not on the strategy s_{-i} of the others. If $c = -\infty$, then the result specializes to [Yu25, Corollary 6.14].

7 Conclusions

We apply lattice-theoretic tools to study games with strategic complementarities. We consider classes of games where the payoff functions may not satisfy the quasisupermodularity nor the single crossing property, but only satisfy “a half” of the properties. Thus, such classes encompass supermodular games, quasisupermodular games and pseudo-supermodular games. We give results on the existence of Nash equilibria, the existence of extremal Nash equilibria, and the order structure of the set of all Nash equilibria. The major results are illustrated with the aid of a number of economics-related examples to which the seminal contributions by [Zho94], [MS94] and [PY17] are not applicable. It is likely to obtain monotone comparative statics for the newly introduced classes of games.

A Proofs

A.1 Proof of Theorem 3.2

In Lemma A.1, we show the existence of best replies for games satisfying (USC). A poset P is called *chain-complete upwards* (resp. *chain-complete downwards*) if for every nonempty chain $C \subset P$, $\sup_P C$ (resp. $\inf_P C$) exists in P . If P is chain-complete upwards and chain-complete downwards, then it is called *chain-complete*, which roughly means that we can take limits of increasing or decreasing sequences in P . A lattice is chain-complete if and only if it is complete, but a chain-complete poset may not be a lattice.

Lemma A.1. *Let $(N, \{S_i\}, \{u_i\})$ be a game satisfying (USC). Then for every $i \in N$, S_i is a complete lattice. For every $i \in N$ and every $s_{-i} \in S_{-i}$, the poset $R_i(s_{-i})$ is nonempty, chain-complete and has maximal and minimal elements.*

Proof. It follows from Lemma B.3 and Zorn’s lemma. $\square$

An essential ingredient in the proof of Theorem 3.2 is Kukushkin’s selection theorem [Kuk13a, Theorem 2.2], which we review. A correspondence $F : X \rightarrow 2^Y$ from a poset X to a lattice Y is called *weakly ascending*, if for any $x < x'$ in X , every $y \in F(x)$ and every $y' \in F(x')$, one has either $y \vee y' \in F(x')$ or $y \wedge y' \in F(x)$. Kukushkin’s theorem states that given a weakly ascending correspondences $F : X \rightarrow 2^Y$, if every value $F(x)$ is nonempty, chain-complete upwards and admits minimal elements, then $F : X \rightarrow 2^Y$ has an increasing selection $f : X \rightarrow Y$.

Now we prove Theorem 3.2. Fix $i \in N$, and we show that $R_i : S_{-i} \rightarrow 2^{S_i}$ is weakly ascending. By symmetry, one may assume that i satisfies Condition 1.

Assume to the contrary that $R_i : S_{-i} \rightarrow 2^{S_i}$ is not weakly ascending. There exist elements $x_{-i} < x'_{-i}$ in S_{-i} , $x_i \in R_i(x_{-i})$ and $x'_i \in R_i(x'_{-i})$ satisfying

$$x_i \wedge x'_i \notin R_i(x_{-i}), \quad x_i \vee x'_i \notin R_i(x'_{-i}).$$

Then $u_i(x_i \wedge x'_i, x_{-i}) < u_i(x_i, x_{-i})$. As $u_i(\cdot, x_{-i})$ meet-superextremal, one has

$$u_i(x_i \vee x'_i, x_{-i}) \geq u_i(x_i, x_{-i}) \wedge u_i(x'_i, x_{-i}) = u_i(x'_i, x_{-i}).$$

As u_i satisfies the upward single-crossing property, this inequality implies $u_i(x_i \vee x'_i, x'_{-i}) \geq u_i(x'_i, x'_{-i})$. This contradicts $x_i \vee x'_i \notin R_i(x'_{-i})$.

By Lemma A.1, the values of $R_i : S_{-i} \rightarrow 2^{S_i}$ are nonempty and chain-complete. Then by [Kuk13a, Theorem 2.2], this weakly ascending correspondence has an increasing selection $r_i : S_{-i} \rightarrow S_i$. For every $s \in S$, let $r(s) = (r_i(s_{-i}))_{i \in N}$ be an element of S whose i -th coordinate is $r_i(s_{-i})$. Thus, $r : S \rightarrow S$ is an increasing selection of the best reply correspondence $R : S \rightarrow 2^S$. By Lemma A.1, S is a complete lattice. Then by Tarski's fixed point theorem [Tar55, Theorem 1], $r : S \rightarrow S$ has a fixed point, which is a Nash equilibrium.

A.2 Proof of Theorem 4.2

Let X, Y be lattices, $F : X \rightarrow 2^Y$ be a correspondence. If for any $x < x'$ in X , every $y \in F(x)$ and every $y' \in F(x')$, there is $u \in F(x')$ with $u \geq y \vee y'$, then we call F *upper C-ascending*. The author [Yu23] gives a fixed point theorem for such correspondences, of which Theorem 4.2 is an application in game theory.

Here is a detailed proof of Theorem 4.2. By symmetry, it suffices to prove the statement without parentheses. We prove that the joint best response correspondence $R : S \rightarrow 2^S$ is upper C-ascending. For this, consider every $i \in N$, any $x_{-i} \leq x'_{-i}$ in S_{-i} , every $y_i \in R_i(x_{-i})$ and every $y'_i \in R_i(x'_{-i})$. Since $u_i(\cdot, x_{-i}) : S_i \rightarrow \mathbb{R}$ satisfies Condition (PS-b), from $u_i(y_i \wedge y'_i, x_{-i}) \leq u_i(y_i, x_{-i})$ one has

$$u_i(y_i \vee y'_i, x_{-i}) \geq u_i(y_i, x_{-i}) \wedge u_i(y'_i, x_{-i}) = u_i(y'_i, x_{-i}).$$

By upward transfer single-crossing property of u_i , there exists $y''_i \in S_i$ with $y''_i \geq y_i \vee y'_i$ and $u_i(y'_i, x'_{-i}) \leq u_i(y''_i, x'_{-i})$. Thus $y''_i \in R_i(x'_{-i})$. Therefore, for any $x \leq x'$ in S , every $y \in R(x)$ and every $y' \in R(x')$, one can choose an element $y'' := (y''_i)_i$ of $R(x')$ with $y'' \geq y \vee y'$.

By Lemma A.1, as the game satisfies (USC), every value of R is nonempty chain-complete. From [Yu23, Lemma A.0.3], an upper C-ascending correspondence with nonempty chain-complete values has a largest fixed point. The largest fixed point of $R : S \rightarrow 2^S$ is the largest Nash equilibrium.

A.3 Proof of Theorem 5.1

In [Top98, p.33], a correspondence $F : X \rightarrow 2^Y$ from a poset X to a lattice Y is called *increasing*, if for any $x < x'$ in X , every $y \in F(x)$ and every $y' \in F(x')$, one has $y \vee y' \in F(x')$ and $y \wedge y' \in F(x)$. Zhou [Zho94, Theorem 1] establishes a fixed point theorem for increasing correspondences whose values are the so-called “closed sublattices”. The author [Yu24] relaxes Zhou’s “closed sublattice” condition to completeness. We shall prove that the best reply correspondence of the game in Theorem 5.1 is increasing. Then we apply our generalization of Zhou’s fixed point theorem, so that we avoid verifying the stronger “closed sublattice” condition.

We give a complete proof. First, we prove that for every $i \in N$, the correspondence $R_i : S_{-i} \rightarrow 2^{S_i}$ is increasing. For this, consider arbitrary $x_{-i} \leq x'_{-i}$, $x_i \in R_i(x_{-i})$ and $x'_i \in R_i(x'_{-i})$. By $x'_i \in R_i(x'_{-i})$, one has

$$u_i(x_i \vee x'_i, x'_{-i}) \leq u_i(x'_i, x'_{-i}). \quad (1)$$

Using Assumption 1 and the single crossing property, we prove $x_i \wedge x'_i \in R_i(x_{-i})$.

- Case 1: for all $s_{-i} \in S_{-i}$, $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ satisfies (PS-a). As u_i satisfies the single crossing property, by (1) one has $u_i(x_i \vee x'_i, x_{-i}) \leq u_i(x'_i, x_{-i})$. Together with $x_i \in R_i(x_{-i})$, it implies

$$u_i(x_i \vee x'_i, x_{-i}) \leq u_i(x_i, x_{-i}) \wedge u_i(x'_i, x_{-i}).$$

As $u_i(\cdot, x_{-i})$ satisfies (PS-a), one has

$$u_i(x_i \wedge x'_i, x_{-i}) \geq u_i(x_i, x_{-i}) \vee u_i(x'_i, x_{-i}) = u_i(x_i, x_{-i})$$

and hence $x_i \wedge x'_i \in R_i(x_{-i})$.

- Case 2: for all $s_{-i} \in S_{-i}$, $u_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ satisfies (DPS-b). From (1) and (DPS-b), one has

$$u_i(x_i \wedge x'_i, x'_{-i}) \geq u_i(x_i, x'_{-i}) \wedge u_i(x'_i, x'_{-i}) = u_i(x_i, x'_{-i}).$$

As u_i satisfies the single crossing property, one has $u_i(x_i \wedge x'_i, x_{-i}) \geq u_i(x_i, x_{-i})$.

Then $x_i \wedge x'_i \in R_i(x_{-i})$.

Similarly, by Assumption 2 and the single crossing property, one has $x_i \vee x'_i \in R_i(x'_{-i})$.

Now the correspondences $R_i : S_{-i} \rightarrow 2^{S_i}$ are increasing, so is $R : S \rightarrow 2^S$. In particular, for every $s \in S$, $R(s)$ is a sublattice of S . By Lemma A.1, $R(s)$ is nonempty and chain-complete. By¹ [Mar76, Corollary 5] (see also [MR90, Footnote 8]), the chain-complete lattice $R(s)$ is a complete lattice. Therefore, from [Yu24, Theorem 1.3], the set of Nash equilibria $\text{Fix}(R)$ is a nonempty complete lattice.

¹A lattice that is chain-complete in our sense is also chain-complete in Markowsky’s sense.

A.4 Proof of Lemma 6.3

Take arbitrary elements $x_i < x'_i$ in S_i and $x_{-i} < x'_{-i}$ in S_{-i} . It suffices to prove that the condition $u_i(x_i, x'_{-i}) \geq u_i(x'_i, x'_{-i})$ implies $u_i(x_i, x_{-i}) \geq u_i(x'_i, x_{-i})$.

First assume $D_i(x_i, x_{-i}) = 0$. By Assumption (D1), one has $D_i(x'_i, x_{-i}) = 0$. Then both $u_i(x_i, x_{-i})$ and $u_i(x'_i, x_{-i})$ equal $-C_i(0)$.

Assume now $D_i(x_i, x_{-i}) > 0$. By Assumption (D1), it implies

$$D_i(x_i, x'_{-i}) > 0, \quad (2)$$

$$\frac{D_i(x'_i, x'_{-i})}{D_i(x_i, x'_{-i})} \leq 1. \quad (3)$$

As $C_i : [0, +\infty) \rightarrow [0, +\infty)$ is a convex function and $x_{-i} < x'_{-i}$, one can choose $c(x_{-i}), c(x'_{-i}), K(x_{-i}), K(x'_{-i}) \in \mathbb{R}$ such that $c(x_{-i}) \leq c(x'_{-i})$ and

$$\begin{aligned} C_i(a) &= c(x_{-i})a + K(x_{-i}), \quad \forall a \in \{D_i(x'_i, x_{-i}), D_i(x_i, x_{-i})\}, \\ C_i(b) &= c(x'_{-i})b + K(x'_{-i}), \quad \forall b \in \{D_i(x'_i, x'_{-i}), D_i(x_i, x'_{-i})\}. \end{aligned}$$

By $u_i(x_i, x'_{-i}) \geq u_i(x'_i, x'_{-i})$, one has

$$(x_i - c(x'_{-i}))D_i(x_i, x'_{-i}) \geq (x'_i - c(x'_{-i}))D_i(x'_i, x'_{-i}) \quad (4)$$

and hence $x'_i - c(x'_{-i}) > x_i - c(x'_{-i}) \geq (x'_i - c(x'_{-i}))D_i(x'_i, x'_{-i})/D_i(x_i, x'_{-i})$. Together with (3), it proves $x'_i - c(x'_{-i}) > 0$. Then one has

$$\frac{x_i - c(x'_{-i})}{x'_i - c(x'_{-i})} \leq \frac{x_i - c(x_{-i})}{x'_i - c(x_{-i})}. \quad (5)$$

Combined with (2) and (4), it implies

$$\frac{D_i(x'_i, x'_{-i})}{D_i(x_i, x'_{-i})} \leq \frac{x_i - c(x'_{-i})}{x'_i - c(x'_{-i})}. \quad (6)$$

By Assumption (D2), one has $D_i(x'_i, x_{-i})/D_i(x_i, x_{-i}) \leq D_i(x'_i, x'_{-i})/D_i(x_i, x'_{-i})$. Combined with (5) and (6), it implies

$$\frac{D_i(x'_i, x_{-i})}{D_i(x_i, x_{-i})} \leq \frac{x_i - c(x_{-i})}{x'_i - c(x_{-i})}.$$

It gives

$$(x'_i - c(x_{-i}))D_i(x'_i, x_{-i}) \leq (x_i - c(x_{-i}))D_i(x_i, x_{-i}),$$

or equivalently $u_i(x'_i, x_{-i}) \leq u_i(x_i, x_{-i})$.

A.5 Proof of Corollary 6.9

For every $i \in N$ and every $s_{-i} \in S_{-i}$, $\underline{u}_i(\cdot, s_{-i}) : S_i \rightarrow \mathbb{R}$ is still upper semicontinuous. Therefore, the new game satisfies (USC). By Lemma 6.7, the function satisfies (PS-b) and (DPS-b). We verify the single crossing property for $\underline{u}_i : S_i \times S_{-i} \rightarrow \mathbb{R}$. Let $s_i < s'_i$ be elements of S_i , and let $s_{-i} < s'_{-i}$ be elements of S_{-i} .

- If $\underline{u}_i(s_i, s_{-i}) < \underline{u}_i(s'_i, s_{-i})$, then $c < \underline{u}_i(s'_i, s_{-i})$. Thus, one has $c < u_i(s'_i, s_{-i})$ and $u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i})$. By assumption, one has $c < u_i(s'_i, s'_{-i})$. As $u_i : S_i \times S_{-i} \rightarrow \mathbb{R}$ satisfies the single crossing property, one has $u_i(s_i, s'_{-i}) < u_i(s'_i, s'_{-i})$ and hence $\underline{u}_i(s_i, s'_{-i}) < u_i(s'_i, s'_{-i}) \leq \underline{u}_i(s'_i, s'_{-i})$.
- If $\underline{u}_i(s_i, s'_{-i}) > \underline{u}_i(s'_i, s'_{-i})$, then $\underline{u}_i(s_i, s'_{-i}) > c$. It implies $u_i(s_i, s'_{-i}) > c$ and $u_i(s_i, s'_{-i}) > u_i(s'_i, s'_{-i})$. By assumption, one has $u_i(s_i, s_{-i}) > c$. As $u_i : S_i \times S_{-i} \rightarrow \mathbb{R}$ satisfies the single crossing property, one has $u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$ and hence $\underline{u}_i(s_i, s_{-i}) \geq u_i(s_i, s_{-i}) > \underline{u}_i(s'_i, s_{-i})$.

Then the result follows from Theorem 5.1.

B Existence of maximum

In a normal form game, each player's optimal response to the strategies chosen by the other players is to maximize his or her own payoff function. In this regard, for a fixed player, the existence of maximum of the corresponding payoff function is equivalent to the existence of best responses. This is how the topological assumption (USC) in Theorems 3.2, 4.2 and 5.1 is used to show that every player always has at least one best reply.

When the strategy sets are subspaces of Euclidean spaces, the Euclidean topology is frequently used in which it is relatively easy to check (USC). For infinite dimensional spaces, however, as Milgrom and Shannon [MS94, Appendix] put it, "it may often be easier to check that a function is upper semi-continuous along chains, and that its domain is a complete lattice, than to determine whether it is continuous and its domain compact in some appropriate topology."

Milgrom and Roberts [MR90, Theorem 2] prove that the maximizer of a *supermodular*, order-theoretically upper semicontinuous function on a complete lattice is a nonempty lattice. Then Milgrom and Shannon [MS94, Theorem A4] generalize the result to *quasisupermodular* functions. These results are analogs of the topological fact (see, e.g., [Ced63, p.991]) that an upper semicontinuous function on a compact space attains its maximum. In Theorem B.1, we give a similar, purely order-theoretical sufficient condition for the existence of maximum.

Theorem B.1 generalizes [Vei92, Ch. 6, Theorem 41], which assumes furthermore that for every $t \in \mathbb{R}$, the set $[f \geq t]$ is *chain-subcomplete* in X . This generalization

is proper, as shown by Example B.2. In [LV92, p.12], a subset S of a lattice X is a *quasisublattice*, if for any $x, y \in S$, at least one of $x \wedge y$ and $x \vee y$ is in S . For a set X , a function $f : X \rightarrow \mathbb{R}$ and $t \in \mathbb{R}$, let $[f \geq t] := \{x \in X \mid f(x) \geq t\}$ be the upper level set.

Theorem B.1. *Let $f : X \rightarrow \mathbb{R}$ be a meet-superextremal function on a nonempty lattice X . Suppose that for every $t \in f(X)$, the set $[f \geq t]$ is chain-complete downwards and admits maximal elements. Then $\arg \max_X f$ is a nonempty quasisublattice of X .*

Proof. Let $T := \{-f(x) \mid x \in X\}$. Define a correspondence

$$F : T \rightarrow 2^X, \quad F(t) = [f \geq -t].$$

By assumption, every value $F(t)$ is chain-complete downwards and admits maximal elements.

We prove that F is weakly ascending. For this, consider any $t < \tau$ in T , every $s \in F(t)$ and every $\sigma \in F(\tau)$. As f is meet-superextremal, one has either $f(s \wedge \sigma) \geq f(s) \geq -t$ or $f(s \vee \sigma) \geq f(\sigma) \vee f(s) \geq -\tau$. Then either $s \wedge \sigma \in F(t)$ or $s \vee \sigma \in F(\tau)$.

Let $\mathcal{P}$ be the set of increasing selections of F . By a dual of [Kuk13a, Theorem 2.2], as F is weakly ascending, $\mathcal{P}$ is nonempty. For $r, r' \in \mathcal{P}$, by $r \leq r'$ we mean $r(t) \leq r'(t)$ for all $t \in T$. Then $(\mathcal{P}, \leq)$ is a poset. We show that every nonempty chain in $\mathcal{P}$ has a lower bound. Let $\{r^i\}_{i \in I}$ be such a chain. For every $t \in T$, the subset $\{r^i(t)\}_{i \in I}$ of $F(t)$ is a chain. Because $F(t)$ is chain-complete downwards, $r(t) := \inf_{F(t)} \{r^i(t)\}_{i \in I}$ exists. For any $t \leq t'$ in T and every $i \in I$, one has $r(t) \leq r^i(t) \leq r^i(t') \leq r(t')$. From $r(t) \in F(t) \subset F(t')$, one has $r(t) \leq \inf_{F(t')} \{r^i(t')\}_{i \in I} = r(t')$. Therefore, $r \in \mathcal{P}$ is a lower bound on the chain $\{r^i\}_i$.

By Zorn's lemma, $\mathcal{P}$ has a minimal element r . For any $t \leq \tau$ in T , one has $r(t) \in F(t) \subset F(\tau)$. Then $\{r(t)\}_{t \leq \tau, t \in T}$ is a nonempty chain in $F(\tau)$. Since $F(\tau)$ is chain-complete downwards, the element $m(\tau) := \inf_{F(\tau)} \{r(t)\}_{t \leq \tau, t \in T}$ exists.

For any $\tau \leq \tau'$ in T , by $F(\tau) \subset F(\tau')$, one has

$$m(\tau) = \inf_{F(\tau)} \{r(t)\}_{t \leq \tau, t \in T} \leq \inf_{F(\tau')} \{r(t)\}_{t \leq \tau, t \in T} = \inf_{F(\tau')} \{r(t)\}_{t \leq \tau', t \in T} = m(\tau').$$

Thus, the map $m : T \rightarrow X$ is increasing and $m \in \mathcal{P}$.

For every $t \in T$, one has $m(t) \leq r(t)$. Consequently, $m \leq r$ in $\mathcal{P}$. Since r is minimal in $\mathcal{P}$, one has $m = r$. Then for any $t' \leq t$ in T , one has $r(t) = m(t) \leq r(t') \leq r(t)$. As a consequence, one has $r(t) = r(t')$. Therefore, there is $s_0 \in X$ such that the map $r : T \rightarrow X$ is constantly s_0 . Then $s_0 \in F(t)$, so $f(s_0) \geq -t$ for all $t \in T$. Therefore, one has $f(s_0) = \max f$. In particular, $\arg \max_X f$ is nonempty.

For any $x, y \in \arg \max_X f$, as f is meet-superextremal, either $f(x \wedge y) \geq f(x) = \max f$ or $f(x \vee y) \leq f(x) \wedge f(y) = \max f$, i.e., either $x \wedge y$ or $x \vee y$ is in $\arg \max_X f$. This shows that $\arg \min_X f$ is a quasisublattice of X . $\square$

Example B.2. Let $f : [0, 2] \rightarrow \mathbb{R}$ be the characteristic function of $[0, 1) \cup \{2\}$. Then f is superextremal. It satisfies the condition of Theorem B.1. However, $[f \geq 1] = [0, 1) \cup \{2\}$ is not chain-subcomplete upwards in $[0, 2]$. In this case, [Vei92, Ch. 6, Theorem 41] is not applicable.

Lemma B.3 shows that the order theoretic assumption in Theorem B.1 is weaker than the following topological assumption involved in (USC):

- there is a compact topology τ on X finer than the interval topology, such that $f : X \rightarrow \mathbb{R}$ is upper semicontinuous in τ .

Lemma B.3. *Let X be a lattice endowed with a compact topology τ finer than the interval topology. Let $f : X \rightarrow \mathbb{R}$ be a function upper semicontinuous in τ . Then X is a complete lattice, and $\arg \max_X f$ is nonempty and chain-complete.*

Proof. The interval topology of X is coarser than a compact topology τ , so it is also compact. Then from the Frink-Birkhoff theorem [Bir73, Thm. 20, p.250], X is a complete lattice. By semicontinuity, for every $t \in \mathbb{R}$, $[f \geq t]$ is closed in (X, τ) . By compactness of (X, τ) , $[f \geq t]$ is compact in τ . The interval topology of $[f \geq t]$ is coarser than the subspace topology of τ , hence also compact. Then by [EW18, Lem. 1], $[f \geq t]$ is chain-complete. By semicontinuity, $\max f$ exists, so $\arg \max_X f = [f \geq \max f]$ is nonempty and chain-complete. $\square$

Remark B.4. Using Theorem B.1, one can replace the topological assumption (USC) in Theorems 3.2, 4.2 and 5.1 by the following order theoretic assumption, without changing the results.

- All S_i are complete lattices. For every $s_{-i} \in S_{-i}$ and every $t \in \mathbb{R}$, every nonempty chain in $[u_i(\cdot, s_{-i}) \geq t]$ has an infimum and an upper bound. (Or dually, for every $s_{-i} \in S_{-i}$ and every $t \in \mathbb{R}$, every nonempty chain in $[u_i(\cdot, s_{-i}) \geq t]$ has a supremum and a lower bound.)

The reason is that we only use (USC) to ensure that the best-reply correspondences R_i have nonempty values. These purely order theoretic results generalize to the case where the payoff functions are chain-valued maps. The variant of Theorem 3.2 thus obtained can be applied to the war of attrition game G^- in [PY17, Example 5].

Remark B.5. In [MR90, p.1261], given a complete lattice X , a function $f : X \rightarrow \mathbb{R}$ is called *order upper semicontinuous* if for every nonempty chain $C \subset X$, one has

$$\limsup_{x \in C, x \uparrow \sup_L(C)} f(x) \leq f(\sup_L C), \quad \limsup_{x \in C, x \downarrow \inf_L(C)} f(x) \leq f(\inf_L C).$$

In that case, for every $t \in f(X)$, $[f \geq t]$ is chain-complete. If f is quasisupermodular and order upper semicontinuous, then Theorem B.1 recovers the existence part of [MS94, Theorem A4]. This generalization is also proper.

We apply Theorem B.1 to optimization of functionals. Let (S, Σ, μ) be a measure space. Let $p \in [1, \infty)$. Let $L^p(\mu)$ be the set of all real-valued μ -measurable functions $g : S \rightarrow \mathbb{R}$ with $\int_S |g|^p d\mu < \infty$. For $f, g \in S$, write $f \leq g$ whenever $f(x) \leq g(x)$ for μ -almost all $x \in S$. Then $(L^p(\mu), \leq)$ is a Riesz space. Equip $L^p(\mu)$ with the L^p -norm topology, so that it is a Banach space.

Corollary B.6. *Let $x \leq y$ be two elements of $L^p(\mu)$. Let $f : [x, y] \rightarrow \mathbb{R}$ be an upper semicontinuous function. If f is meet-superextremal or join-superextremal, then it attains its maximum.*

Proof. We prove the case with meet-superextremality, and the other case follows by symmetry. By [Fre10, 244L, Theorem], $[x, y]$ is a complete lattice. For every $t \in f([x, y])$, we prove that $[f \geq t]$ is chain-complete. Let $C = \{z_\alpha\}_{\alpha \in I}$ be a nonempty chain in $[f \geq t]$. Since $z_\alpha \uparrow \sup C$, one has order convergence $z_\alpha \xrightarrow{o} \sup C$ in the sense of [AB06, p.33]. From [Fre10, 244Y (e)], order convergence in $L^p(\mu)$ implies topological convergence, so $z_\alpha \xrightarrow{L^p} \sup C$. As f is upper semicontinuous, $[f \geq t]$ is closed in $L^p(\mu)$ and hence $\sup C \in [f \geq t]$. Similarly, one has $\inf C \in [f \geq t]$. Then by Theorem B.1, as f is meet-superextremal, $\arg \max f$ is nonempty. $\square$

Remark B.7. In Corollary B.6, $[x, y]$ may not be compact in the L^p -norm topology. More generally, using [AB06, Corollary 4.10], one can replace $L^p(\mu)$ by an arbitrary Banach lattice with order continuous norm.

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